

Population Projections That Can Be Wrong for a Reason

Research design and preliminary backtest for a mechanism-explicit framework to 2150

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Analysis protocol and preliminary evidence, version 0.1.0-draft
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Draft status. The historical backtest and deterministic projection architecture are implemented. The UW probabilistic baseline is being assembled. Mechanism estimates and long-run comparative results do not yet exist and are not implied by this draft.

Abstract

Conditional on age-specific fertility, mortality, and migration schedules, cohort-component projection is deterministic. Long-run results nevertheless depend on uncertain structural assumptions about how those schedules evolve. This research protocol develops a reproducible framework that keeps those two layers separate. It first backtests observable portions of archived United Nations population projection vintages against WPP 2024 estimates. It then validates a single-year cohort-component engine against published WPP 2024 projection inputs and outputs. The next stage will pass probabilistic fertility and mortality trajectories through that same engine before comparing explicit accounts of post-transition fertility. The intended contribution is not a single preferred population total for 2150. It is a framework in which competing explanations use identical demographic bookkeeping, expose their assumptions, preserve dated predictions, and identify nearer-term outcomes by which they can later be rejected or revised. Probabilistic and mechanism-comparison results are forthcoming and are not claimed here.

1 Introduction

Population projections are often discussed as if disagreement centered on the arithmetic. At long horizons, it does not. Once age-specific fertility, mortality, and migration rates have been supplied, cohort-component projection is deterministic bookkeeping. The scientifically consequential uncertainty is upstream: what processes generate those rates, especially after fertility has fallen below replacement.

This distinction matters more as the horizon lengthens. By 2150, nearly everyone alive will have been born after the 2024 base year. Small but persistent differences in fertility therefore compound through both cohort size and later births. Mortality and migration remain important, but neither removes the need to state why long-run fertility should stabilize, rebound, or continue to fall.

The United Nations now produces deterministic and probabilistic population projections using a cohort-component engine and Bayesian trajectories for the components of demographic change [United Nations Department of Economic and Social Affairs, Population Division, 2024a]. The

underlying probabilistic framework is well established [Alkema et al., 2011, Raftery et al., 2012, Raftery and Ševčíková, 2023]. The contribution is not a new cohort-component engine, but a transparent comparison of alternative rate-generating models through a common validated accounting layer, with predictions preserved in a form that can be scored later.

Three questions will organize the completed study:

1. What signed and regional error patterns recur in observable portions of past World Population Prospects (WPP) projection vintages?
2. How much do long-run outcomes change when alternative fertility mechanisms are passed through identical demographic accounting?
3. Which nearer-term cohort outcomes distinguish those mechanisms soon enough to provide a meaningful test?

The final question is central. Because a 2150 endpoint cannot be evaluated within a useful horizon, the study pre-specifies nearer-term cohort outcomes that distinguish the competing mechanisms. The workflow therefore treats prior predictive checking, retrospective testing, and future scorekeeping as part of the research design rather than presentation added afterward [Gelman and Shalizi, 2013, Gneiting and Raftery, 2007].

2 Historical projection backtest

The retrospective analysis uses archived WPP revisions whose target years have since elapsed and are covered by a later WPP estimate series. Each vintage is evaluated only over horizons that had resolved by the comparison date. The current implementation covers the 1992, 1994, 1996, 2000, 2002, 2004, 2006, and 2008 revisions and reports results at the world and stable regional levels, avoiding country comparisons where changing borders would create false errors. The omitted 1998 release and later revisions will be documented explicitly when the historical archive is extended; the present eight-vintage set is the implemented and reproducible scope rather than a claim to cover every release in the period [United Nations Department of Economic and Social Affairs, Population Division, 2024b].

The comparison series is WPP 2024. It is an updated demographic estimate, not ground truth. This distinction is especially important for weak-data countries and much less consequential for world totals. The low and high projection variants are fertility scenarios rather than probability intervals; whether a later estimate falls between them is therefore described as inclusion in the printed range, not probabilistic calibration. The current calculation conditions on the WPP 2024 estimates as fixed comparison values and does not propagate uncertainty in those estimates.

For population, the analysis records signed and absolute relative error by vintage, target year, horizon, and region. Fertility and life expectancy are evaluated separately to identify which projected components moved in the same direction as population error. Overlapping target years and successive vintages are not independent observations. The final paper will therefore present vintage-level distributions and robustness analyses rather than treat every vintage-year pair as an independent replication.

Across the scored world-level vintage-year pairs, the mean signed population error is -2.45%, where a negative value means that the projection was below the later WPP 2024 estimate. The 1992 revision projected too high; every included revision from 1996 onward projected too low. The forecasts overlap in target years and are not independent tests of an unchanged model.

Thirty years of UN world population projections, marked

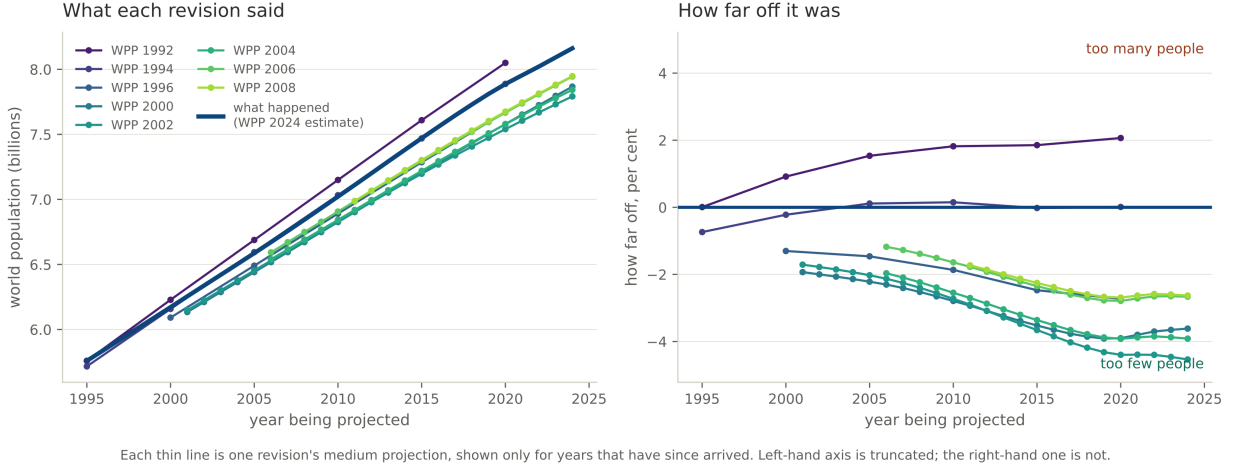


Figure 1: Medium-variant world population projections from eight archived WPP revisions, shown only for target years available in the later WPP 2024 estimate series. The right panel reports signed relative error; negative values are underprojections.

The component backtest points in different directions across regions. Mean African fertility error is -9.8%, while mean Eastern Asian fertility error is +14.9%. Life expectancy is lower than the later comparison series by -1.32 years on average. These figures identify where the archived trajectories missed; they do not establish a causal mechanism. In a separate range check, 41 of 117 (35%) of the world-level observations fall inside the printed low-to-high fertility variants. That quantity is not a probability-interval coverage rate.

A formal component decomposition and extension to later gradeable revisions remain required before making stronger causal claims.

3 Projection architecture

3.1 A deterministic accounting layer

The population state is indexed by country, sex, and single year of age from zero through an open-ended 100-plus group. Counts refer to 1 January. One model step advances the state to the following 1 January by applying survival into each age, computing births from mid-year female exposure and age-specific fertility, splitting births by sex ratio, and applying net migration.

For location l , sex s , and closed age a , the aging operation is schematically

$$N_{l,s,a,t+1} = S_{l,s,a,t}N_{l,s,a-1,t} + M_{l,s,a,t}, \quad (1)$$

with separate treatment of births and the open age group. Births are computed from fertility rates $f_{l,a,t}$ and the average of beginning- and end-of-year female exposure. Childbearing ages span 10–54 so the implementation retains the nonzero rates reported in the published WPP inputs outside ages 15–49.

The two fertility mistakes that drove the population misses

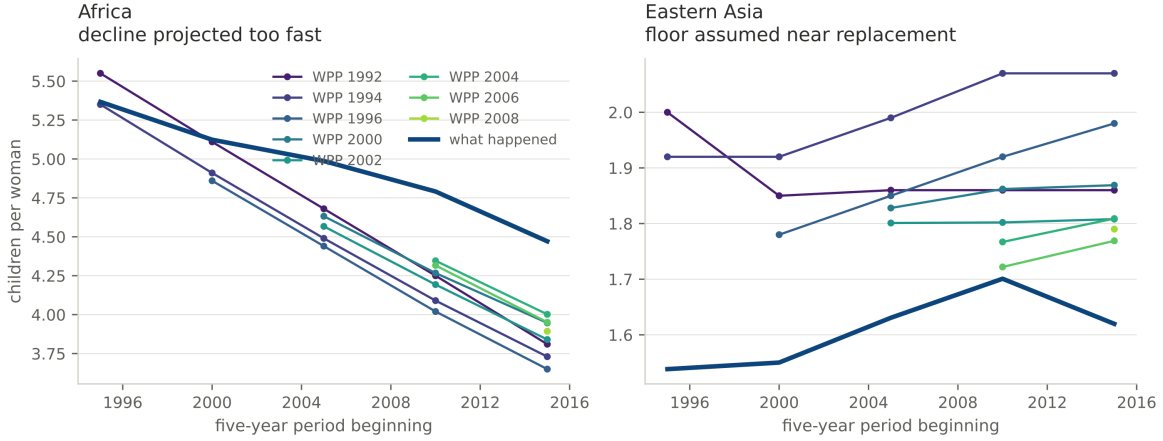


Figure 2: Archived total-fertility projections for Africa and Eastern Asia compared with WPP 2024 estimates. The opposite-signed misses motivate a mechanism-explicit comparison, but do not themselves attribute cause.

This module contains no account of future fertility. That is a design constraint: if a behavioral assumption enters the accounting layer, later results can no longer be attributed cleanly to the mechanism that generated them.

3.2 Implementation validation

The engine is checked against WPP 2024 results using published inputs rather than fitted corrections. The primary validation uses zero migration because all required terms are independently available. A medium-path migration term backed out of the UN trajectory is retained only as a diagnostic: it absorbs differences between implementations and is not independent evidence.

Validation is performed at world, country, age-group, open-age, and constant-fertility levels. Passing these checks establishes implementation fidelity to the selected accounting conventions. It does not validate any substantive assumption about future fertility or mortality.

The zero-migration comparison is the primary independent-input accounting test. The constant-fertility comparison is a secondary check and inherits the derived migration residual used to reproduce the corresponding WPP variant.

3.3 Two-stage probabilistic propagation

The sampler and the population engine remain separate. Statistical models or source adapters first produce complete draws of demographic rates. Each draw is then advanced through the deterministic engine one at a time. This follows the general architecture of probabilistic population projection [Raftery et al., 2012, Ševčíková et al., 2016] while keeping the deterministic installation lightweight and the provenance of each component visible.

Table 1: Selected deterministic engine checks at 2100. Errors compare the independent implementation with the corresponding published WPP 2024 result.

Check	Absolute relative error
World population, zero migration	0.001%
Worst country above 10,000 people	0.07%
Worst closed five-year age-sex group	0.006%
Open age group, 100-plus	1.08%
World population, constant fertility	0.05%

4 Probabilistic baseline

Analysis pending. A genuine Finland fixture has been extracted through the official University of Washington accessors and reconciled. This section will be completed only after bulk extraction and conversion into age-specific schedules.

The baseline will begin from 1,000 annual UW trajectories of total fertility and paired female and male life expectancy aligned to WPP 2024. These are UW products, not official UN posterior draws. Their native accessors can apply stored alignment shifts, so selected binary files cannot be read in isolation without risking a silent change in values [University of Washington Center for Statistics and the Social Sciences, 2024].

The source boundary preserves total fertility and life expectancy at full precision together with their original trajectory identities. A separately versioned schedule converter then expands total fertility into age-specific fertility and life expectancy into age- and sex-specific survival. Conversion is accepted only if the detailed schedules reconstruct the compact source quantities within stated tolerances.

Female and male life expectancy trajectories are paired by UW’s joint model. The fertility and mortality archives are separate products. If equal ordinal pairing is used across those products, it will be a reproducible project assumption and will not be described as a joint fertility-mortality posterior.

The UW archives contain 236 locations while WPP 2024 contains 237. The extracted code sets confirm that the omitted location is Holy See (M49 336); no posterior trajectory is invented to fill it. The source also contains a 2023 anchor; the project preserves that value for validation and begins projected rate draws in 2024.

These source trajectories end in 2100. Any projection from 2101 through 2150 therefore requires separately named extension policies for fertility, mortality, and migration. The present download covers fertility and mortality; the resulting ensemble will be conditional on an explicit migration assumption, base population, and sex ratio at birth rather than being described as a complete probabilistic WPP population projection.

5 Planned mechanism comparison

Analysis pending. No mechanism has been fitted and no mechanism result is claimed in this draft. This section states the comparison the completed paper must make.

The mechanism layer separates two questions that are easily conflated:

1. **Composition:** which fertility-related dispositions or social types become more common across generations?
2. **Environment:** how many births does a person of a given type realize under changing economic, institutional, and cultural conditions?

Intergenerational persistence can shift composition while development, opportunity cost, partnership patterns, and institutions shift the mapping from disposition to realized fertility. A completed model must therefore allow selection and environment to move simultaneously and in opposite directions. It must also distinguish genetic transmission, family or cultural transmission, group retention, intermarriage, and assimilation rather than compress those channels into one uninterpretable coefficient.

Every mechanism parameter will be accompanied by a provenance table stating its independent evidence source, the outcome used to estimate it, its prior, and whether it remains a scenario choice. A parameter may not be fitted to the same outcome that it is later said to explain. Period total fertility may be reported descriptively, but it will not be used as the scoring target because changes in birth timing can distort it. Completed-cohort fertility and parity progression are needed to distinguish tempo from quantum [Bongaarts and Feeney, 1998].

6 Model checking and scorekeeping

The Bayesian workflow will begin with prior predictive simulation. Priors that violate stated demographic constraints or assign substantial mass to extreme paths unsupported by an explicit scenario rationale will be revised before fitting. Posterior predictive checks will assess whether the model reproduces salient features of the observed data and whether predictive failures remain concentrated by region or forecast horizon.

Retrospective evaluation will use rolling information cutoffs. Genuine vintage-pure hindcasts will use only contemporaneously available inputs; reconstructed pseudo-out-of-sample exercises will be labelled separately and will not be conflated with posterior predictive checks. Parameters are estimated only from information available before the target period, and predictions are evaluated on later cohort outcomes. Probabilistic comparisons will use proper scoring rules, including the continuous ranked probability score already implemented in the project [Gneiting and Raftery, 2007]. Calibration and sharpness will be reported separately.

Future predictions will be written as immutable vintages. Each commitment will identify the quantity, units, resolution source, expected resolution date, stored predictive distribution, and scoring rule. A new model may supersede an old one, but it will not overwrite what the old model predicted.

7 Discussion

The present framework is designed to expose structural disagreement rather than make long-run uncertainty look smaller than it is. That transparency also clarifies several limitations.

First, WPP estimates are model-based comparison data, particularly where census and registration systems are weak. Second, country-level paths to 2150 can be dominated by migration assumptions

that are less stable than world totals. Third, converting total fertility and life expectancy into age schedules adds uncertainty that should be tested rather than hidden. Fourth, social-group behavior may change as groups grow, so persistence estimated from small groups cannot be extrapolated mechanically. Finally, a detailed mechanism can mislead more effectively than a transparent trend model if its parameters are not independently constrained.

Mortality deserves similar restraint. [Oeppen and Vaupel \[2002\]](#) document sustained increases in record female life expectancy despite repeatedly proposed ceilings, but that observation does not identify one correct age-specific mortality path through 2150. Long-run results will therefore remain conditional on named mortality and extension policies.

The interactive webpage is intended to become a public companion to the analysis, not evidence in itself. It should make assumptions and uncertainty explorable while the paper provides durable methods, limitations, and, after release, citable results.

8 Conclusion

Long-run population projection is most useful when it can be wrong for a stated reason. This project fixes and validates a demographic accounting layer, reports a preliminary historical backtest, and establishes an immutable prediction format. The probabilistic baseline and mechanism comparison remain under construction. The completed paper will be successful not if it identifies a dramatic population total for 2150, but if it makes competing mechanisms comparable and commits them to nearer-term outcomes that can change our confidence in those mechanisms.

A Reproducibility and provenance

The repository separates source acquisition, ingestion, deterministic projection, statistical modeling, and presentation. Download manifests record source URLs, versions, sizes, and SHA-256 fingerprints. Unmatched location codes stop a build. Generated model outputs remain distinct from reference targets used for validation.

The final artifact build will create every paper and webpage figure from the same result tables. The paper will record its version, Git revision, WPP data vintage, projection vintage, source fingerprints, software environment, and generation date. A reviewed PDF will be stored at a stable path, while released versions remain immutable.

The first real UW fixture used R 4.4.2, `bayesTFR` 7.4-4, and the exact UW Git revision of `bayesLife` 5.3-0 used to create the archive. Official accessors exported Finland’s 1,000 trajectories for 2023–2100 and applied 78 stored alignment-shift values to each component. The extracted code set confirmed 236 locations and identified Holy See (M49 336) as the sole WPP 2024 omission. Archive, reader-source, and export-file SHA-256 fingerprints are recorded in the repository manifests.

The schedule-conversion appendix will report reconstruction tolerances and the fertility-mortality pairing rule. The mechanism appendix will provide the complete parameter-provenance table, prior predictive checks, sampler diagnostics, retrospective tests, and sensitivity analyses.

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